



Faculty of Engineering

Department of Biomedical Engineering

Configuration List for The Non-Contact Clinical Thermometer

Part Number: 01-MIY-004

Rev-1

Date: 18/12/2025



CHANGE RECORDS

FIRST RELEASE

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REVISIONS

Rev. No	Change Description	Date	QA Approval

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1. PURPOSE

The purpose of the configuration item list for the non-contact type thermometer is to formally identify, document, and control all hardware and electronic components required for the development, assembly, and maintenance of the device. By listing each component with its corresponding part number and item description, the configuration item list ensures traceability, consistency, and accuracy throughout the product lifecycle. This documentation supports effective procurement, quality assurance, and change management by providing a single, authoritative reference that aligns design specifications with the actual components used in the thermometer.

2. SCOPE

The scope of the configuration item list covers all materials and components necessary for the non-contact thermometer, including sensing elements, electronic modules, mechanical parts, power components, and interconnections. For each item, the list provides the relevant part number, a clear description, and reference links to approved suppliers for purchasing the materials. The scope is limited to components that directly contribute to the functionality, assembly, and verification of the thermometer, and it serves as a reference for all engineers during design and manufacturing activities.

3. CONFIGURATION ITEM LIST

Table 1: Configuration Item List for Non Contact Thermometer

Item No	Part Number	Description
1	01-PRT-001	MCU Kit, ESP32 C3 Mini
2	01-PRT-002	MLX90614, Infrared Camera Sensor
3	01-PRT-003	TP4056, Battery Charger Module
4	01-PRT-004	Display Kit, 0.96" LCD Display
5	01-PRT-005	On/Off Slider Switch
6	01-PRT-006	Measurement Button
7	01-PRT-007	Buzzer
8	01-PRT-008	Resistor
9	01-PRT-009	JST Connector

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10	01-PRT-010	Lithium Ion Battery
11	01-PRT-011	Sensor PCB
12	01-PRT-012	Housing Assembly

4. REFERENCES

1. Microcontroller – ESP 32 C3 super mini - <https://www.direnc.net/esp32-c3-mini-wifi-bluetooth-modulu>
2. Infrared camera sensor - MLX90614 - https://www.robotistan.com/gy906-mlx90614-kizilotesi-sensoru-modulu?language=tr&h=e5094714&sgm_campaign=fcs_c12e1a8d78000&sgm_source=9896%7Cproduct&sgm_action=search&sgm_term=mlx90614&sgm_pinned=false
3. Battery charger – TP4056 - <https://www.robotistan.com/tp4056-type-c-1s-37v-lipo-li-ion-pil-sarj-devresi>
4. Battery and buzzer connector – JST connector - https://www.robotistan.com/ds1066-2mrw6xb-2-pin-90-derece-pcb-jst-konnektor?language=tr&h=4bd8387c&sgm_campaign=fcs_c12e1a8d78000&sgm_source=12491%7Cproduct&sgm_action=search&sgm_term=jst&sgm_pinned=falseii
5. LCD Display - <https://www.direnc.net/128x64-096-4-pin-oled-grafik-lcd-ekran-ssd1306-i2c-beyaz>
6. ON/OFF Slider Switch - <https://www.direnc.net/surgulu-on-off-switch-180>
7. Begin Measurement Button - <https://www.direnc.net/6x6-05mm-tach-buton-4-bacak>
8. Resistor – 1kohm, 10kohm - [Buy 0R 5% 1W 2512 SMD Resistor at a Good Price - Direnc.net®](#)
9. Active Buzzer - [3-24V Circuit 22.6mm Ear Buzzer - Buy Black at a Reasonable Price - Direnc.net®](#)

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